Extra-Curricular Activity Project Assignment

Automatic Generation and Analysis of Signal Flow Graphs

Project Overview

In this project, you will design and implement a software tool that automatically converts a block diagram representation of a control system into its corresponding signal flow graph (SFG) and then computes the overall system transfer function using Mason's Gain Formula.

Learning Objectives

By completing this project, you will be able to:

- Interpret and model control system block diagrams computationally.
- Represent control systems as signal flow graphs programmatically.
- Implement path-finding and loop-detection algorithms to identify forward paths and loops in an SFG.
- Apply Mason's Gain Formula to calculate the overall transfer function.
- Generate a graphical visualization of the SFG, helping users understand system structure and dependencies.
- Integrate control theory with software engineering principles, including data structures, algorithm design, and user interface development.